# INNER PRODUCTS, NORMS, BRAS, KETS, AND PROJECTION

A detailed mathematical and quantum-computing reference

From geometric overlap to the Born rule, projectors, expectation values, unitary preservation, and multi-qubit inner products.

*Topic 4 in the Quantum Mathematics Foundation series*

# How to use this volume

This chapter is designed as a long-term reference rather than a quick lecture. It begins with real-vector geometry, builds the complex inner product carefully, and then connects each construction to quantum measurement. Work through the examples with paper and pencil; use the exercises to test whether the notation has become operational.

Prerequisites: complex numbers, vectors and vector spaces, matrices, conjugate transpose, and basic qubit notation.

## Learning objectives

- Construct bras from kets and compute complex inner products correctly.
- Use inner products to define norms, distances, angles, orthogonality, and coordinate extraction.
- Calculate measurement amplitudes and probabilities using the Born rule.
- Distinguish inner products, outer products, and projector operators.
- Understand completeness, expectation values, variance, unitary preservation, and Gram-Schmidt orthonormalization.
- Extend all definitions to multi-qubit state spaces.

## Contents

# 1. From dot products to inner products

For real vectors, the dot product compresses two vectors into one scalar. It measures directional agreement and provides the algebra behind length, angle, projection, and orthogonality.

$$\text{For } u = (u1, u2, \ldots, un)^T \text{ and } v = (v1, v2, \ldots, vn)^T:$$
$$u \cdot v = u1\, v1 + u2\, v2 + \ldots + un\, vn$$

## Geometrically,

$$u \cdot v = ||u||\ ||v|| \cos(\theta)$$

Therefore, a positive dot product means the vectors point partly in the same direction, a negative value indicates an opposing component, and zero means they are perpendicular.

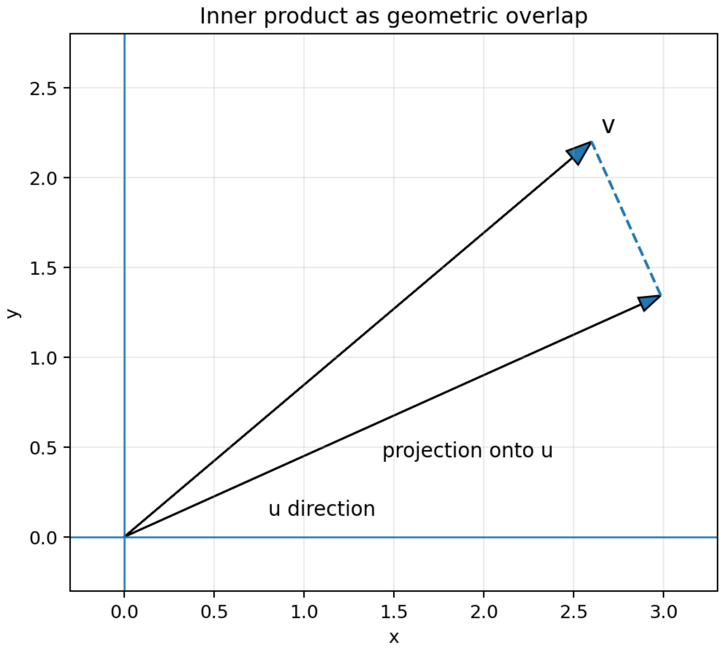


*Figure 1. The dot product determines the scalar projection of one vector along another.*

### Example

Let u = (1,2)^T and v = (3,4)^T. Then u . v = 1(3)+2(4)=11. Their norms are sqrt(5) and 5, so cos(theta)=11/(5 sqrt(5)).

## Why this formula is insufficient for quantum vectors

Quantum state vectors contain complex numbers. If we copied the real formula without conjugation, a nonzero vector could have zero or even complex squared length. For example, (1,i)^T would give 1^2+i^2=0. A length cannot behave this way.

The correction is to multiply each component of the first vector by its complex conjugate. This converts terms such as i into |i|^2=1 and makes self-overlap real and non-negative.

# 2. Complex inner products and bra-ket notation

For complex vectors |phi> and |psi>, the inner product is defined by conjugating the first vector and then performing the usual row-by-column multiplication.

<phi|psi> = phi1* psi1 + phi2* psi2 + ... + phin* psin

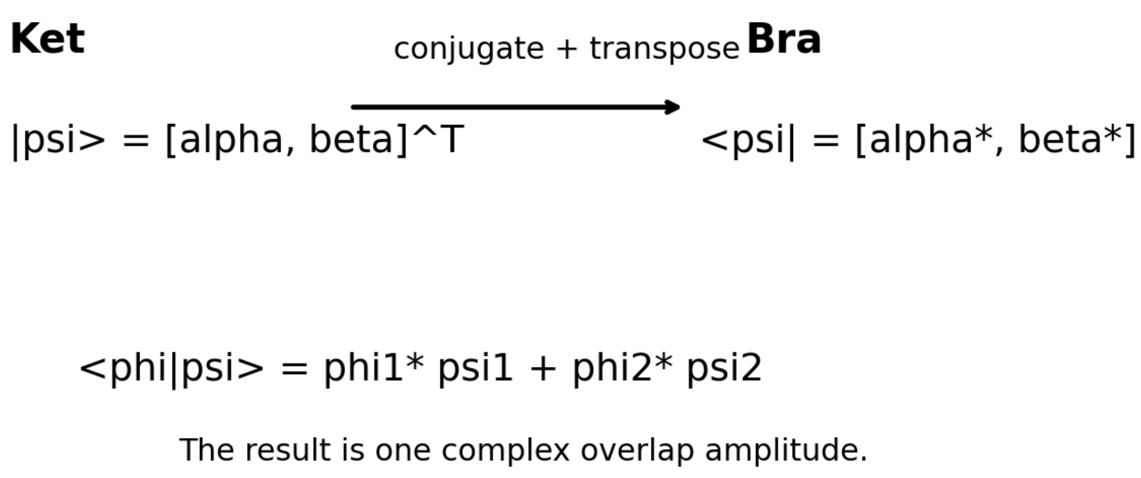


*Figure 2. The bra is the conjugate transpose of the ket.*

### If

|psi> = (alpha, beta)^T

then its bra is

<psi| = (alpha*, beta*)

and the self-inner-product becomes

<psi|psi> = |alpha|^2 + |beta|^2.

## Notation and dimensions

| Object | Shape | Example | Result type |
|---|---|---|---|
| Ket |psi> | n x 1 column | (alpha,beta)^T | vector |
| Bra <psi| | 1 x n row | (alpha*,beta*) | covector |
| Inner product <phi|psi> | (1 x n)(n x 1) | sum phi_k* psi_k | scalar |
| Outer product |phi><psi| | (n x 1)(1 x n) | matrix of phi_j psi_k* | operator |

## Worked example

Let |phi>=(1,i)^T and |psi>=(i,1)^T. Then <phi|=(1,-i), so <phi|psi>=1(i)+(-i)(1)=0. The vectors are orthogonal.

## Conjugate symmetry

<phi|psi> = (<psi|phi>)*

The order of a complex inner product cannot normally be reversed without taking a conjugate. This distinction disappears only when the overlap is real.

# 3. Algebraic properties of inner products

The following rules are used constantly in quantum derivations.

## Linearity in the ket

<phi|(a|psi> + b|chi>) = a<phi|psi> + b<phi|chi>

## Conjugate linearity in the bra

(a|phi> + b|chi>)^dagger |psi> = a* <phi|psi> + b* <chi|psi>

## Positive definiteness

<psi|psi> >= 0, with equality only for |psi> = 0.

## Conjugate symmetry

<phi|psi> = <psi|phi>*

These properties are not arbitrary conventions. They are precisely what is required for the inner product to behave like generalized geometry in a complex vector space.

## Moving scalars through bras and kets

If a scalar multiplies a ket, it remains unchanged when pulled out of the inner product. If it multiplies the vector that becomes the bra, it is conjugated.

<phi| (c|psi>) = c <phi|psi>
<c phi|psi> = c* <phi|psi>

### Example with phase

Let |phi>=i|0>. Then <phi|=-i<0|. Hence <phi|0>=-i, not i. This is a common source of sign errors.

## Sesquilinearity

An inner product is linear in one argument and conjugate-linear in the other. This combined behavior is called sesquilinearity. Physics usually adopts the convention used here: conjugate-linear in the bra and linear in the ket.

# 4. Norm, distance, angle, and normalization

The inner product generates the norm:

$$||psi|| = sqrt(<psi|psi>)$$

For $|psi>=(alpha,beta)^T$:

$$||psi|| = sqrt(|alpha|^2 + |beta|^2).$$

A pure quantum state must be normalized:

$$<psi|psi> = 1.$$

## Normalization procedure

For any nonzero vector $|v>$, divide by its norm:

$$|v\_normalized> = |v> / ||v||.$$

### Example

For $|v>=(1+i,2)^T$, the squared norm is $|1+i|^2+|2|^2=2+4=6$. Thus $|v\_normalized>=(1/sqrt(6))(1+i,2)^T$.

## Distance

$$d(|phi>,|psi>) = || |phi> - |psi> ||.$$

This is the ordinary Euclidean distance generalized to complex coordinates. In quantum information, other state-distance measures are often more physically meaningful because global phase must be ignored, but norm distance remains mathematically fundamental.

## Angle and overlap

For normalized real vectors, the dot product equals cos(theta). For complex vectors, the overlap can have a phase. Its magnitude still quantifies closeness:

$$0 <= |<phi|psi>| <= 1.$$

Two normalized vectors have maximum overlap when they differ only by a global phase, and zero overlap when they are orthogonal.

## Global phase

$$|phi> = e^(i gamma)|psi> => |<phi|psi>|^2 = 1.$$

Thus physical pure states are rays: normalized vectors identified up to a common unit complex factor.

# 5. Orthogonality and orthonormal bases

Two vectors are orthogonal when their inner product vanishes:

$$\langle \phi|\psi\rangle = 0.$$

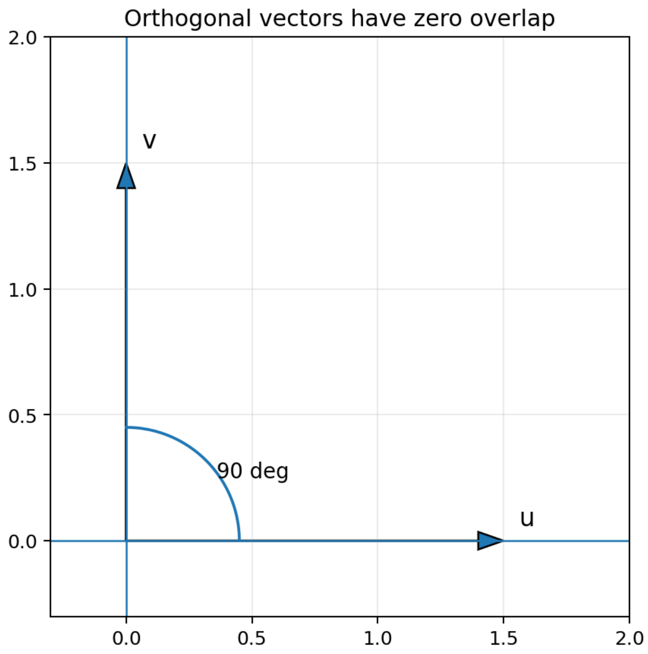


*Figure 3. Zero inner product generalizes perpendicularity.*

Orthogonal quantum states can be perfectly distinguished by a suitable projective measurement. Nonorthogonal pure states cannot be perfectly distinguished from a single copy.

## Orthonormal sets

A set $\{|e_k\rangle\}$ is orthonormal if

$$\langle e_j|e_k\rangle = \delta_{jk}.$$

Here $\delta_{jk}$ is 1 when $j=k$ and 0 otherwise. Each basis state has unit norm, and different basis states are orthogonal.

### Computational basis

$$|0\rangle=(1,0)^T,\ |1\rangle=(0,1)^T;$$
$$\langle 0|0\rangle=\langle 1|1\rangle=1,\ \langle 0|1\rangle=0.$$

### X basis

$$|+\rangle=(|0\rangle+|1\rangle)/\sqrt{2},\ |-\rangle=(|0\rangle-|1\rangle)/\sqrt{2}.$$

The X-basis vectors are also orthonormal. The same vector space can have infinitely many orthonormal bases.

## Coordinate extraction

If $\{|e_k\rangle\}$ is an orthonormal basis and

$$|\psi\rangle = \sum_k c_k |e_k\rangle,$$

then multiplying by $\langle e_j|$ gives

$$c_j = \langle e_j|\psi\rangle.$$

The inner product extracts the coordinate of $|\psi\rangle$ along a chosen basis direction.

# 6. Measurement amplitudes and the Born rule

Suppose a projective measurement uses the orthonormal basis {|m_k>}. The amplitude for outcome k is

$$a_k = \langle m_k|\psi\rangle.$$

The Born rule converts this amplitude into a probability:

$$p(k) = |\langle m_k|\psi\rangle|^2.$$

The modulus square is essential: the amplitude may be complex, but the probability must be real and non-negative.

## Computational-basis measurement

For a qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

we have

$$\langle 0|\psi\rangle=\alpha,\ \langle 1|\psi\rangle=\beta,$$
$$p(0)=|\alpha|^2,\ p(1)=|\beta|^2.$$

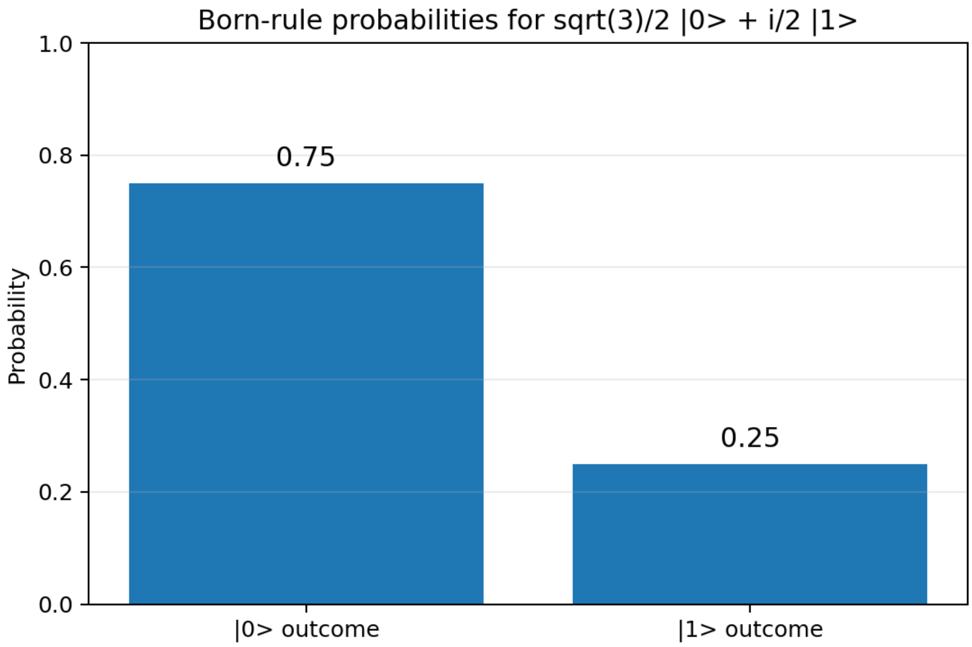


*Figure 4. Theoretical probabilities for a state with amplitudes sqrt(3)/2 and i/2.*

## Normalization guarantees total probability one

$$\sum_k p(k) = \sum_k |\langle m_k|\psi\rangle|^2 = 1.$$

## Amplitude is not probability

The overlap <m_k|psi> contains phase information. Squaring its magnitude produces the probability, but the phase can affect later interference before measurement.

### Example

For $|\psi\rangle=(\sqrt{3}/2)|0\rangle+(i/2)|1\rangle$, the probabilities are 3/4 and 1/4. The factor i does not change the second probability, but it changes the state relative to one with amplitude 1/2.

# 7. Measurement in an alternative basis

A state is not intrinsically random or deterministic without specifying the measurement basis. Measurement asks how much the state overlaps each vector in the chosen basis.

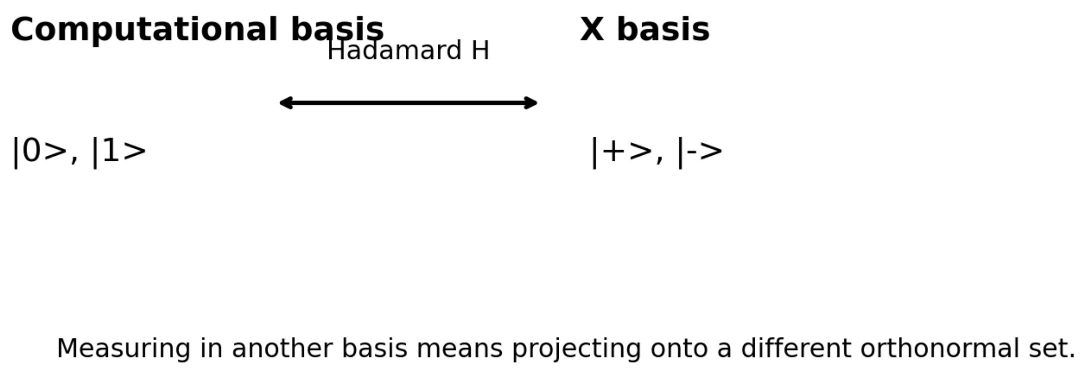


*Figure 5. The Hadamard connects computational and X-basis descriptions.*

## Measuring |0> in the X basis

<+|0> = 1/sqrt(2), <-|0> = 1/sqrt(2).

Therefore:

p(+) = 1/2, p(-) = 1/2.

The same state |0> gives a definite answer in the Z basis and equal probabilities in the X basis.

## Measuring a general qubit in the X basis

For |psi>=alpha|0>+beta|1>:

<+|psi> = (alpha+beta)/sqrt(2),
<-|psi> = (alpha-beta)/sqrt(2).

Hence

p(+) = |alpha+beta|^2/2, p(-) = |alpha-beta|^2/2.

These formulas show interference directly: relative phase affects addition and subtraction even when |alpha|^2 and |beta|^2 are unchanged.

## Circuit implementation

To measure in the X basis using hardware that measures only in the computational basis, apply H before measurement. H maps |+> to |0> and |-> to |1>.

# 8. Outer products

The inner product <phi|psi> produces a scalar. Reversing the order produces an operator:

$$|phi><psi|$$

For

$$|phi>=(a,b)^T \text{ and } <psi|=(c^*,d^*),$$

the outer product is

$$|phi><psi| = [[a c^*, a d^*], [b c^*, b d^*]].$$

Outer products are building blocks for projectors, density matrices, spectral decompositions, and quantum channels.

## Example: |0><1|

$$|0><1| = [[0,1],[0,0]].$$

Its action is

$$|0><1| (alpha|0>+beta|1>) = beta|0>.$$

The bra <1| extracts the |1> amplitude; the ket |0> replaces the direction with |0>.

## Rank-one operators

An outer product |phi><psi| maps every input into a scalar multiple of |phi>. Its image therefore lies in a one-dimensional subspace, making it a rank-one operator when both vectors are nonzero.

## Adjoint

$$(|phi><psi|)^{dagger} = |psi><phi|.$$

A projector |phi><phi| is therefore Hermitian when |phi> is normalized.

# 9. Projectors and state components

For a normalized vector |phi>, the orthogonal projector onto its direction is

P_phi = |phi><phi|.

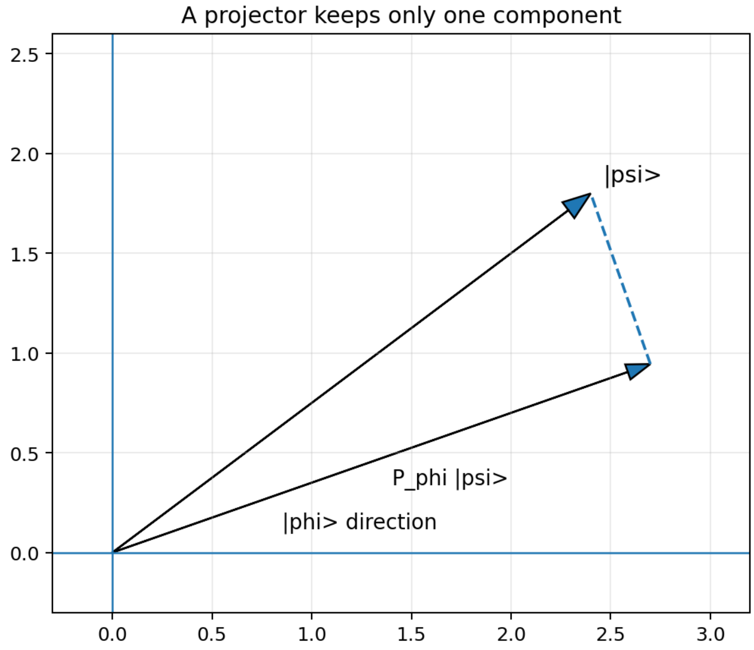


*Figure 6. Projection removes the component perpendicular to |phi>.*

## Action on a state

P_phi|psi> = |phi><phi|psi>.

The overlap <phi|psi> gives the coefficient, and |phi> supplies the direction.

## Projector properties

P_phi^2 = P_phi, P_phi^dagger = P_phi.

Idempotence means that after a vector has been projected, projecting it again changes nothing.

## Eigenvalues

If P|v>=lambda|v>, then P^2=P implies lambda^2=lambda. Therefore:

Projector eigenvalues are 0 or 1.

Vectors inside the target subspace have eigenvalue 1; vectors orthogonal to it have eigenvalue 0.

## Measurement probability through projectors

p_phi = <psi|P_phi|psi> = |<phi|psi>|^2.

This is the Born rule written in operator form.

## Post-measurement state

Conditioned on obtaining the outcome associated with projector P_k, the state becomes

|psi_after> = P_k|psi> / sqrt(<psi|P_k|psi>).

The denominator renormalizes the selected component.

# 10. Completeness and resolution of the identity

For an orthonormal basis $\{|e_k\rangle\}$, the projectors sum to the identity:

$$\mathrm{sum}_k |e_k\rangle\langle e_k| = I.$$

For a one-qubit computational basis:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = [[1,0],[0,1]] = I.$$

## Coordinate reconstruction

Insert the identity before a state:

$$|psi\rangle = I|psi\rangle = \mathrm{sum}_k |e_k\rangle\langle e_k|psi\rangle.$$

Therefore:

$$|psi\rangle = \mathrm{sum}_k \langle e_k|psi\rangle\, |e_k\rangle.$$

This formula states that a vector is reconstructed by adding its projections onto all orthonormal basis directions.

## Parseval identity

Taking the norm of a normalized state gives:

$$\mathrm{sum}_k |\langle e_k|psi\rangle|^2 = 1.$$

The total squared overlap with a complete orthonormal basis equals the squared norm of the vector. In quantum mechanics, this is the statement that measurement probabilities sum to one.

## Projective measurement conditions

A collection $\{P_k\}$ defines an ideal projective measurement when:

- $P_k = P_k^\dagger$ and $P_k^2=P_k$.
- $P_j P_k=0$ for $j \neq k$.
- $\mathrm{sum}_k P_k=I$.

These conditions guarantee mutually exclusive outcomes that jointly exhaust the state space.

# 11. Expectation values and uncertainty

For a Hermitian observable A, the expectation value in state |psi> is

<A>_psi = <psi|A|psi>.

If A has spectral decomposition

A = sum_k a_k |a_k><a_k|,

then

<A> = sum_k a_k |<a_k|psi>|^2.

This is exactly the probability-weighted average of possible eigenvalues.

## Example: Pauli Z

For |psi>=alpha|0>+beta|1> and Z=|0><0|-|1><1|:

<Z> = |alpha|^2 - |beta|^2.

Each shot returns +1 or -1. The expectation value is the long-run average, not necessarily an individual outcome.

## Variance

(Delta A)^2 = <A^2> - <A>^2.

Variance measures the spread of measurement results. An eigenstate of A has zero variance because repeated measurements always produce the same eigenvalue.

### Example for Pauli observables

Because Z^2=I, <Z^2>=1 for every normalized state. Thus:

(Delta Z)^2 = 1 - <Z>^2.

For |+>, <Z>=0 and Delta Z=1. For |0>, <Z>=1 and Delta Z=0.

# 12. Unitary transformations preserve inner products

A matrix U is unitary when U^dagger U=I. If |psi'>=U|psi> and |phi'>=U|phi>, then

<phi'|psi'> = <phi|U^dagger U|psi> = <phi|psi>.

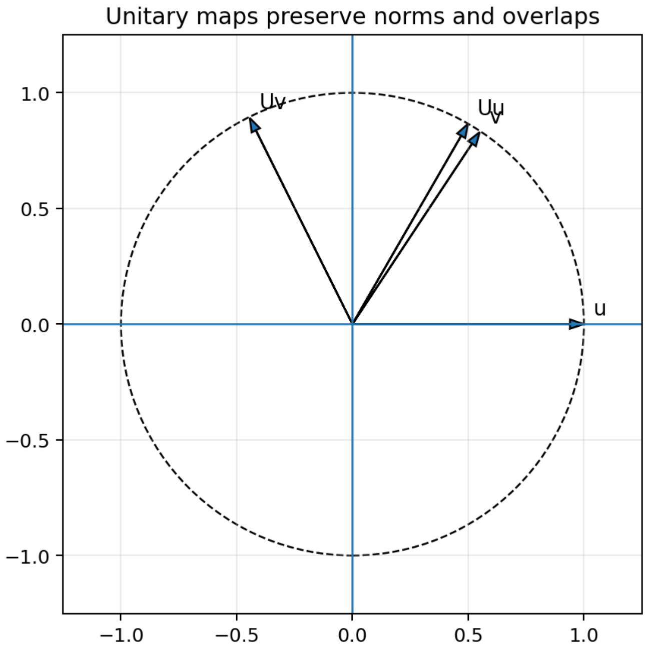


*Figure 7. A unitary transformation acts like a generalized rotation: geometry is preserved.*

## Consequences

- Norms are preserved: ||U|psi>||=||psi||.
- Orthogonality is preserved: if <phi|psi>=0, then <Uphi|Upsi>=0.
- Pairwise distinguishability and transition probabilities are preserved.
- Unitary evolution is reversible, with U^-1=U^dagger.

## Quantum-gate interpretation

Closed-system quantum gates are unitary because they must map normalized states to normalized states without destroying information. Gates may redistribute amplitudes and change phases, but total probability and inner-product geometry remain intact.

### Example: Hadamard

H maps the orthonormal computational basis {|0>,|1>} to the orthonormal X basis {|+>,|->}. Since the images remain orthonormal, the columns of H are orthonormal and H is unitary.

# 13. Cauchy-Schwarz and triangle inequalities

The Cauchy-Schwarz inequality states:

$$|\langle \phi|\psi\rangle| \le \|\phi\|\,\|\psi\|.$$

For normalized states:

$$|\langle \phi|\psi\rangle| \le 1.$$

Equality holds exactly when the vectors are linearly dependent. For normalized quantum states, this means they differ only by global phase.

## Why it matters

- It guarantees Born probabilities never exceed one.
- It bounds overlaps and transition amplitudes.
- It underlies geometric definitions of angles and state fidelity.
- It proves that the norm generated by the inner product is mathematically consistent.

## Proof idea

For nonzero |phi>, consider the component of |psi> orthogonal to |phi>:

$$|r\rangle = |\psi\rangle - (\langle \phi|\psi\rangle/\langle \phi|\phi\rangle)\,|\phi\rangle.$$

Because <r|r> is non-negative, expanding it produces the inequality.

## Triangle inequality

$$\|\phi+\psi\| \le \|\phi\| + \|\psi\|.$$

The length of a sum is no greater than the sum of the individual lengths. Together with positive definiteness and homogeneity, this makes the inner-product norm a valid norm.

## Pythagorean theorem

When <phi|psi>=0:

$$\|\phi+\psi\|^2 = \|\phi\|^2 + \|\psi\|^2.$$

The familiar real-plane theorem survives unchanged in complex Hilbert space when orthogonality is defined through the inner product.

# 14. Gram-Schmidt orthonormalization

A linearly independent set need not be orthogonal. Gram-Schmidt converts it into an orthonormal set spanning the same subspace.

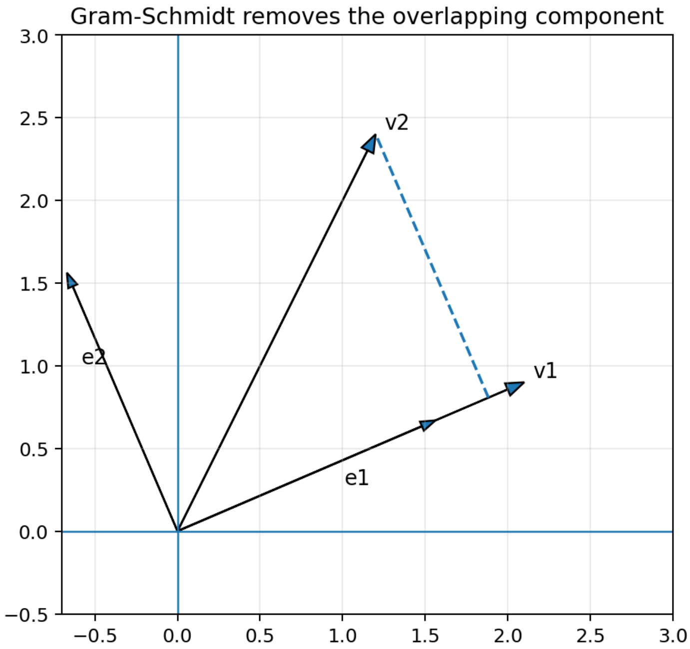


*Figure 8. Subtracting the projection removes overlap with the first direction.*

## Two-vector procedure

Start with independent vectors |v1> and |v2>.

1. Normalize the first vector: |e1> = |v1>/||v1||.
2. Remove from |v2> the component along |e1>: |w2> = |v2> - |e1><e1|v2>.
3. Normalize the remainder: |e2> = |w2>/||w2||.

Then <e1|e2>=0, both vectors have norm one, and span{e1,e2}=span{v1,v2}.

## Complex example

Let |v1>=(1,i)^T and |v2>=(1,1)^T. First, |e1>=(1/sqrt(2))(1,i)^T. The coefficient <e1|v2>=(1-i)/sqrt(2). Subtracting that projection leaves a vector orthogonal to |e1>, which can then be normalized.

## Quantum relevance

Orthonormal bases are required for standard projective measurements. Gram-Schmidt also appears when constructing bases for subspaces, deriving QR decompositions, and stabilizing numerical algorithms.

# 15. Multi-qubit inner products

For two qubits, a general state has four amplitudes:

|psi> = alpha00|00> + alpha01|01> + alpha10|10> + alpha11|11>.


*Figure 9. The computational basis of two qubits has four orthonormal vectors.*

For another state |phi>, the inner product is

<phi|psi> = phi00* alpha00 + phi01* alpha01 + phi10* alpha10 + phi11* alpha11.

The definition is unchanged; only the number of components grows.

## Tensor-product factorization

For product states:

<phi1 tensor phi2 | psi1 tensor psi2> = <phi1|psi1> <phi2|psi2>.

Thus overlaps of separable states factor into overlaps of their subsystems.

## Computational-basis orthonormality

<xy|uv> = delta_xu delta_yv.

For n qubits, different bit strings label mutually orthogonal basis states.

## Bell-state example

|Phi+>=(|00>+|11>)/sqrt(2), |Phi->=(|00>-|11>)/sqrt(2).

Their overlap is zero, so they are orthogonal and can be perfectly distinguished by a suitable joint measurement.

# 16. Worked examples

## Example 1: bra and norm

Given |psi>=(1+i, 2-i)^T, the bra is <psi|=(1-i,2+i). The squared norm is |1+i|^2+|2-i|^2=2+5=7, so ||psi||=sqrt(7).

## Example 2: overlap and probability

Let |phi>=(1/sqrt(2))(1,1)^T and |psi>=(1/sqrt(2))(1,i)^T. Then

<phi|psi> = (1+i)/2, |<phi|psi>|^2 = 1/2.

A measurement containing |phi> returns that outcome with probability 1/2.

## Example 3: projector matrix

For |+>=(1/sqrt(2))(1,1)^T:

P_+ = |+><+| = (1/2)[[1,1],[1,1]].

Applying P_+ to |0> gives (1/2)(1,1)^T=(1/sqrt(2))|+>. Its squared norm is 1/2, which is the probability of the + outcome.

## Example 4: expectation value

For |psi>=cos(theta/2)|0>+e^(i phi)sin(theta/2)|1>:

<Z> = cos^2(theta/2)-sin^2(theta/2)=cos(theta).

## Example 5: unitary preservation

If H maps |0> to |+> and |1> to |->, then <+|->=<0|H^dagger H|1>=<0|1>=0. The gate preserves orthogonality.

## Example 6: state normalization after projection

Suppose |psi>=sqrt(3)/2|0>+1/2|1>. If the P_0 outcome occurs, the unnormalized component is sqrt(3)/2|0>. The outcome probability is 3/4, and dividing by sqrt(3/4)=sqrt(3)/2 gives the final state |0>.

# 17. Common mistakes and diagnostic checks

| Mistake | Why it is wrong | Reliable check |
|---|---|---|
| Using transpose instead of conjugate transpose | Complex phases are not conjugated, so norms and probabilities become incorrect. | For \|psi>=(1,i), self-overlap must be 2, not 0. |
| Treating <phi\|psi> as a probability | It is generally complex and carries phase. | Take \|<phi\|psi>\|^2. |
| Confusing <phi\|psi> with \|phi><psi\| | The first is a scalar; the second is a matrix. | Check dimensions before multiplying. |
| Assuming same basis probabilities mean same state | Relative phases can differ. | Test another measurement basis or compute overlap |
| Forgetting to renormalize after projection | The projected component usually has norm sqrt(p), not 1. | Divide by the square root of the outcome probability |
| Reversing inner-product order freely | The result is conjugated when order is reversed. | Use <phi\|psi>=<psi\|phi>*. |
| Assuming nonorthogonal states are perfectly distinguishable | Their measurement distributions necessarily overlap. | Perfect discrimination requires zero inner product. |

## Fast dimensional check

(1 x n)(n x 1) = scalar; (n x 1)(1 x n) = n x n matrix.

## Fast physical check

- Probabilities must be real and lie between 0 and 1.
- A normalized state must have self-inner-product 1.
- A complete basis measurement must have probabilities summing to 1.
- Unitary transformations cannot change overlaps or norms.

# 18. Exercises

## A. Bras, inner products, and norms

1. Write the bra corresponding to |psi>=(2-i, 1+3i)^T.
2. Calculate <phi|psi> for |phi>=(1,i)^T and |psi>=(2,-i)^T.
3. Find the norm of |v>=(1+i,2, -i)^T.
4. Normalize |v>=(1-i,2i)^T.
5. Prove directly that <psi|psi> is always real and non-negative.

## B. Orthogonality and bases

1. Determine whether (1,i)^T and (1,-i)^T are orthogonal.
2. Verify that |+> and |-> form an orthonormal basis.
3. Express |1> in the X basis.
4. Given normalized |phi> and |psi>, show that |<phi|psi>|=1 implies they differ only by a global phase.
5. Use Gram-Schmidt to orthonormalize (1,0,1)^T and (1,1,0)^T.

## C. Measurement and projectors

1. For |psi>=(1/2)|0>+(sqrt(3)/2)|1>, find Z-basis probabilities.
2. For the same state, find X-basis probabilities.
3. Construct P_- = |-><-| as a matrix.
4. Apply P_- to |0>, find the outcome probability, and normalize the post-measurement state.
5. Show that P_+ + P_- = I and P_+P_-=0.

## D. Expectation and multi-qubit overlap

1. Compute <Z> and Delta Z for |psi>=(sqrt(3)/2)|0>+(1/2)|1>.
2. Compute <X> for |psi>=(|0>+i|1>)/sqrt(2).
3. Find the overlap between (|00>+|11>)/sqrt(2) and (|00>-|11>)/sqrt(2).
4. Prove <phi1 tensor phi2|psi1 tensor psi2>=<phi1|psi1><phi2|psi2>.
5. A unitary U maps two vectors |a>,|b> to |a'>,|b'>. Prove that their transition probability is unchanged.

# 19. Selected solutions

## 1. Bra construction

|psi>=(2-i,1+3i)^T => <psi|=(2+i,1-3i).

## 2. Inner product

<phi|=(1,-i). Thus <phi|psi>=1(2)+(-i)(-i)=2-1=1.

## 3. Norm

The squared norm is |1+i|^2+|2|^2+|-i|^2=2+4+1=7. Therefore ||v||=sqrt(7).

## 4. Normalization

For |v>=(1-i,2i)^T, ||v||^2=2+4=6. Hence |v_normalized>=(1/sqrt(6))(1-i,2i)^T.

## 6. Orthogonality check

Let u=(1,i)^T and v=(1,-i)^T. Then <u|=(1,-i), so <u|v>=1+(-i)(-i)=1-1=0.

## 8. Express |1> in X basis

|1>=(|+>-|->)/sqrt(2).

## 10. Gram-Schmidt

Let v1=(1,0,1)^T. Then e1=(1/sqrt(2))(1,0,1)^T. The overlap <e1|v2>=1/sqrt(2). Subtracting gives w2=(1/2,1,-1/2)^T, whose norm is sqrt(3/2). Therefore e2=(1/sqrt(6))(1,2,-1)^T.

## 12. X-basis probabilities

For alpha=1/2 and beta=sqrt(3)/2:

p(+)=(2+sqrt(3))/4, p(-)=(2-sqrt(3))/4.

## 13. Projector

P_-=(1/2)[[1,-1],[-1,1]].

## 14. Projection of |0>

P_-|0>=(1/2)(1,-1)^T=(1/sqrt(2))|->. The probability is 1/2, and the normalized post-measurement state is |->.

## 16. Expectation and uncertainty

<Z>=3/4-1/4=1/2. Since Z^2=I, (Delta Z)^2=1-(1/2)^2=3/4, so Delta Z=sqrt(3)/2.

## 17. X expectation

For |psi>=(1,i)^T/sqrt(2), X|psi>=(i,1)^T/sqrt(2). The inner product is (1,-i).(i,1)/2=(i-i)/2=0.

## 18. Bell-state overlap

<Phi+|Phi->=(1/2)(1-1)=0.

# 20. Compact reference sheet

<table>
<tr><th>Concept</th><th>Formula / condition</th><th>Quantum meaning</th></tr>
<tr><td>Bra</td><td><psi|=(|psi>)^dagger</td><td>Conjugate-transposed state vector</td></tr>
<tr><td>Inner product</td><td><phi|psi>=sum phi_k* psi_k</td><td>Complex overlap amplitude</td></tr>
<tr><td>Norm</td><td>||psi||=sqrt(<psi|psi>)</td><td>State-vector length</td></tr>
<tr><td>Normalization</td><td><psi|psi>=1</td><td>Total probability is one</td></tr>
<tr><td>Orthogonality</td><td><phi|psi>=0</td><td>Perfect distinguishability in a suitable basis</td></tr>
<tr><td>Born rule</td><td>p(phi)=|<phi|psi>|^2</td><td>Probability of outcome |phi></td></tr>
<tr><td>Outer product</td><td>|phi><psi|</td><td>Rank-one linear operator</td></tr>
<tr><td>Projector</td><td>P_phi=|phi><phi|</td><td>Selects the |phi> component</td></tr>
<tr><td>Projector probability</td><td><psi|P_phi|psi></td><td>Born rule in operator form</td></tr>
<tr><td>Completeness</td><td>sum_k |e_k><e_k|=I</td><td>Basis projectors cover the space</td></tr>
<tr><td>Expectation</td><td><A>=<psi|A|psi></td><td>Average measurement result</td></tr>
<tr><td>Variance</td><td>(Delta A)^2=<A^2>-<A>^2</td><td>Measurement uncertainty</td></tr>
<tr><td>Unitary invariance</td><td><Uphi|Upsi>=<phi|psi></td><td>Quantum gates preserve geometry</td></tr>
<tr><td>Cauchy-Schwarz</td><td>|<phi|psi>|<=||phi||||psi||</td><td>Bounds overlaps and probabilities</td></tr>
</table>

## Essential mental model

Inner product = overlap. Norm = self-overlap. Measurement amplitude = overlap with an outcome state. Probability = squared magnitude of that overlap. Projector = operator that extracts the overlapping component.

## Next topic

Eigenvalues and eigenvectors: definite measurement outcomes, eigenspaces, diagonalization, spectral decomposition, and Pauli eigenbases.

*End of Topic 4*